Lab 01	Get Started
DS604 Introduction to Data Management, Autumn'2026; pm_jat @ daiict	

In this lab, the first task you do is to set up PgAdmin4 for accessing the PostgreSQL server from your system. Then you give a beginning to real work.

(1) Learn to connect with PostgreSQL server. Refer to the document
http://intranet.daiict.ac.in/~pm_jat/ConnectingToPostgreSQLServer.pdf
for instructions for the same!

(2) Look at the following pages and understand the basics of creating tables and using CREATE TABLE statement -
https://www.postgresql.org/docs/10/ddl-basics.html
https://www.postgresql.org/docs/10/ddl-constraints.html

(3) Look at the following pages and learn about basic data types supported by PostgreSQL
https://www.postgresql.org/docs/10/datatype.html
(Look at only 8.1.1, 8.1.2, 8.1.3, 8.3, 8.5.1)

Connect to PostgreSQL through PgAdmin4 as described in the document mentioned in point (1)
Your TA should be able to help you here.

(4) Create Schema named Company using following command at SQL prompt-

```
CREATE SCHEMA XIT;
```

(5) Set XIT as "work schema" (also referred as "default schema") by issuing the following command:

```
SET SEARCH_PATH TO XIT;
```

(6) Create table DEPARTMENT by running following command. This table is to store details of all departments the XIT has.

```
CREATE TABLE department (
    did character(2) NOT NULL,
    dname character varying(30) NOT NULL,
    PRIMARY KEY (did)
);
```

This command creates a table in your database and XIT schema. The metadata of the table is stored in the system catalog. PostgreSQL calls it "Information Schema". We can query information schema about the table. Type the following and observe the output.

```
SELECT table_name, column_name, data_type
     FROM information_schema.columns
     WHERE table_name = 'department';
```

(7) Put some data into Department Table using following statements

```
--INSERT into department
INSERT INTO department
    VALUES ('CS', 'Computer Engineering');
INSERT INTO department
    VALUES ('EE', 'Electrical Engineering');
INSERT INTO department
    VALUES ('ME', 'Mechanical Engineering');
INSERT INTO department
    VALUES ('CE', 'Civil Engineering');
```

(8) List all departments from the Department table

```
SELECT * FROM DEPARTMENT;
```

(9) Create table PROGRAM by running following command. This table is to store details of all programs the XIT offers. It also keeps "reference" to the department under which the program is offered.

```
CREATE TABLE program (
    pid character(3) NOT NULL,
    pname character varying(20) NOT NULL,
    intake smallint,
    did character(2),
    PRIMARY KEY (pid),
    FOREIGN KEY (did) REFERENCES department(did)
        ON UPDATE CASCADE ON DELETE SET NULL
);
```

(10) Put some data into the Program Table using following statements

```
--INSERT into program
INSERT INTO program VALUES ('BCS', 'BTech(CS)', 60, 'CS');
INSERT INTO program VALUES ('MCS', 'MTech(CS)', 20, 'CS');
INSERT INTO program VALUES ('BEE', 'BTech(EE)', 40, 'EE');
INSERT INTO program VALUES ('BME', 'BTech(ME)', 40, 'ME');
INSERT INTO program VALUES ('BIT', 'BTech(IT)', 40, 'CS');
INSERT INTO program VALUES ('BEC', 'BTech(EC)', 40, 'EE');
```

(11) List all programs from the Program table

```
SELECT * FROM program;
```

(12) Create table STUDENT by running following command. This table is to store details of all students that are studying at XIT. It also keeps "reference" to program in which the student is studying

```
CREATE TABLE student (
    studid smallint NOT NULL,
    name character varying(20) NOT NULL,
    progid character(3) ,
    batch smallint,
    cpi numeric(2,1),
  PRIMARY KEY (studid),
  FOREIGN KEY (progid) REFERENCES program(pid)
            ON UPDATE CASCADE ON DELETE CASCADE
);
```

(13) Put some data in student table

```
--INSERT into Student
INSERT INTO student VALUES (1011,'Shally','BCS',2010,8.0);
INSERT INTO student VALUES (1018,'Jitendra','BEE',2010,8.0);
INSERT INTO student VALUES (1110,'Rohit','BCS',2011,7.9);
INSERT INTO student VALUES (1113,'Pharhan','BEE',2011,8.0);
INSERT INTO student VALUES (1118,'Aesha','BEE',2011,8.0);
INSERT INTO student VALUES (1120,'Manav','BEC',2011,7.9);
INSERT INTO student VALUES (1215,'Mehul','BIT',2012,7.9);
INSERT INTO student VALUES (1218,'Megh','BEE',2012,8.0);
INSERT INTO student VALUES (1219,'Misri','MCS',2012,6.8);
INSERT INTO student VALUES (1312,'Amita','MCS',2013,7.0);
INSERT INTO student VALUES (1401,'Kirit','MCS',2014,1.0);
INSERT INTO student VALUES (1402,'Mukesh','MCS',2014,NULL);
```

(14) List all students

```
SELECT * FROM student;
```

(15) Create another schema "company" and make that as new work schema

```
CREATE SCHEMA COMPANY;
SET SEARCH_PATH TO COMPANY;
```

(16) DDL script for creating tables of company database is given in Appendix A at the end of this document

(17) Add some tuples to the company database by executing statements in the script. http://intranet.daiict.ac.in/~pm_jat/insert_company.sql

(18) List down all tables in COMPANY schema

(19) Print structures of all tables in the company database using "DESCRIBE table" command.

Annexure-A

```
CREATE TABLE employee (
      fname VARCHAR(20),
      minit CHAR(1),
      lname VARCHAR(20),
      ssn DECIMAL(9,0),
      bdate DATE,
      address VARCHAR(30),
      sex CHAR(1),
      salary DECIMAL(10,0),
      superssn DECIMAL(10,0),
      dno SMALLINT,
      PRIMARY KEY (ssn),
      FOREIGN KEY (superssn) REFERENCES employee(ssn)
            ON DELETE SET DEFAULT ON UPDATE CASCADE
);

CREATE TABLE department (
      dname VARCHAR(20),
      dno SMALLINT,
      mgrssn DECIMAL(9,0),
      mgrstartdate DATE,
      PRIMARY KEY (dno),
      FOREIGN KEY (mgrssn) REFERENCES employee(ssn)
            ON DELETE SET DEFAULT ON UPDATE CASCADE);

CREATE TABLE dept_locations(
      dno SMALLINT,
      dlocation VARCHAR(20),
      FOREIGN KEY (dno) REFERENCES department on delete cascade,
      PRIMARY KEY (dno,dlocation)
);

CREATE TABLE project(
      pname VARCHAR(20),
      pno SMALLINT,
      plocation VARCHAR(20),
      dno SMALLINT,
      PRIMARY KEY (pno),
      FOREIGN KEY (dno) REFERENCES department(dno)
);

CREATE TABLE works_on(
      essn DECIMAL(9,0),
      pno SMALLINT,
      hours DECIMAL(5,1),
      FOREIGN KEY (essn) REFERENCES employee(ssn)
            ON DELETE CASCADE ON UPDATE CASCADE,
      FOREIGN KEY (pno) REFERENCES project(pno)
            ON DELETE CASCADE ON UPDATE CASCADE,
      PRIMARY KEY (essn,pno)
);
```

```
CREATE TABLE dependent(
      essn DECIMAL(9,0),
      dependent_name VARCHAR(20),
      sex CHAR(1),
      bdate DATE,
      relationship VARCHAR(20),
      FOREIGN KEY (essn) REFERENCES employee(ssn)
            ON DELETE CASCADE ON UPDATE CASCADE,
      PRIMARY KEY (essn,dependent_name)
);

ALTER table employee add FOREIGN KEY (dno) REFERENCES
department(dno);
```